📊 Retail Sales & Customer Insights Analysis

1. 📌 Project Overview

This project focuses on analyzing retail transaction data to understand customer behavior, identify high-value customers, and generate business insights. The analysis was performed using Python, SQL, and Power BI.

2. 🎯 Objectives

- Analyze overall sales performance
- Identify top customers, products, and countries
- Understand revenue trends over time
- Segment customers using RFM analysis
- Build an interactive dashboard for decision-making

3. 🛠 Tools & Technologies

- Python (Pandas, NumPy)
- SQL (MySQL)
- Power BI
- Jupyter Notebook

4. 📂 Dataset Description

The dataset contains retail transaction records with the following fields:

- Invoice Number
- Stock Code
- Description
- Quantity
- Invoice Date
- Unit Price
- Customer ID
- Country

- Total Price

5. 🧹 Data Cleaning (Python)

Performed the following steps:

- Removed duplicate records
- Handled missing values
- Converted data types (dates, numeric fields)
- Created a new column:
    - **Total Price = Quantity × Unit Price**
- Ensured consistency in categorical fields

6. 🧠 Feature Engineering

Created additional features:

- Total Price (transaction value)
- Reference Date (latest transaction date)
- RFM metrics:
    - Recency → Days since last purchase
    - Frequency → Number of purchases
    - Monetary → Total spending

7. 📊 SQL Analysis

Performed key business queries:

- Total Revenue
- Total Customers
- Total Orders
- Revenue by Country
- Monthly Revenue Trends
- Top Customers by Revenue
- Most Sold Products

- Average Order Value

8. 📈 RFM Customer Segmentation

Customers were segmented based on behavior:

- **High Value** → Recent + Frequent buyers
- **Regular** → Moderate activity
- **At Risk** → Inactive customers

Segmentation logic:

- Recency ≤ 30 and Frequency ≥ 5 → High Value
- Recency > 90 → At Risk
- Others → Regular

9. 📊 Power BI Dashboard

Two dashboards were created:

🔹 **Page 1: Sales Overview**

- Total Revenue, Customers, Orders, Avg Order Value
- Revenue by Country
- Monthly Revenue Trends
- Top Customers
- Top Products

🔹 **Page 2: Customer Segmentation**

- Customer distribution by segment
- Revenue contribution by segment
- Average revenue per customer

10. 🔍 Key Insights

- United Kingdom contributes the highest revenue
- A small number of customers generate a large portion of revenue
- Certain products dominate sales volume

- High-value customers contribute the majority of revenue
- At-risk customers show declining engagement

11. 💡 Business Recommendations

- Focus on retaining high-value customers with loyalty programs
- Target at-risk customers with personalized offers
- Promote top-performing products
- Expand business in high-revenue countries
- Optimize inventory based on demand trends

12. 🚀 Conclusion

This project demonstrates how data can be transformed into actionable insights using Python, SQL, and Power BI. The dashboard provides a clear understanding of customer behavior and sales performance, helping businesses make informed decisions.